Effect of frequency-to-place mismatch and frequency warp on speech and music sound quality in acoustic cochlear implant simulation

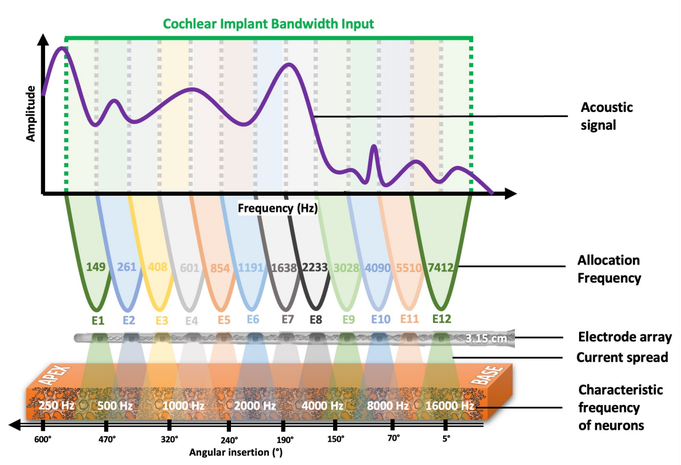
Louis Villejoubert[1], Lorenzo Picinali[2], Kathleen Faulkner[3], Deborah Vickers[1]

[1] Sound Lab, Cambridge Hearing Group, Department of Clinical Neuroscience, Cambridge University, UK
[2] Dyson School of Design Engineering, Imperial College London, London, UK
[3] Oticon, Ballerup, Denmark

Introduction

In more recent years a greater emphasis has been placed on improving the **sound quality** for individuals using cochlear implants (CIs). Indeed, many post-lingually deafened adults with CIs have speech recognition performance close to normal hearing (NH) listeners in quiet environments, but yet complain about poor sound quality, especially for music. The contribution of CI technology to the degradation of music quality has been well documented, but little is known about the relative importance of different aspects relating to the technology and how they interact with the personal preferences and hearing characteristics of the individual. One individual factor that is suspected to contribute largely to the sound quality degradation is the **frequency-to-place mismatch** (FTPM) induced by the **incomplete insertion of the electrode array** inside the cochlea.

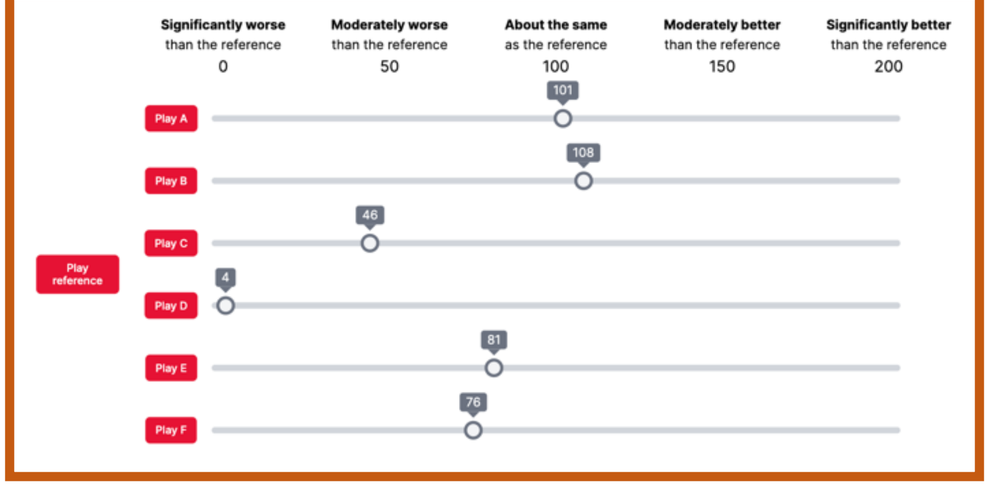
Frequency-to-place mismatch illustration



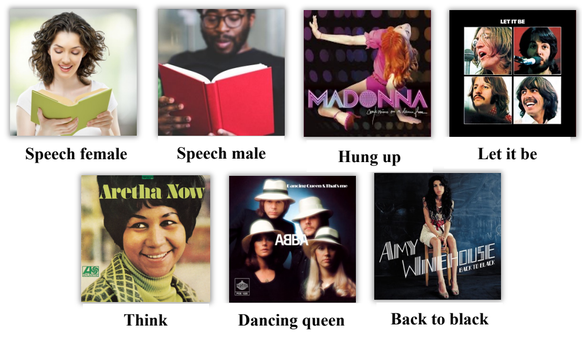
FTPM is defined as a deviation between the **characteristic frequency** and the **allocation frequency** assigned during CI fitting. Whereas much of the speech perception improvement occurs during the first months after CI activation, recent studies have shown that a significant FTPM could negatively affect this adaptation process. Alongside the improvements in speech perception over time from device activation patients often also change their reports of sound quality, from “high pitched”, “screechy”, “dalek-like” to more “normal” or “natural” after extensive listening experience. However, the contributions of the electrode array position and FTPM on sound quality are not clear.

This research investigates the link between sound quality and FTPM in acute listening situations with CI simulations and NH listeners (experiment 1) and how pre-modifications of speech and music material can be used to explore and understand the impact of FTPM (experiment 2).

MUSHRA Method



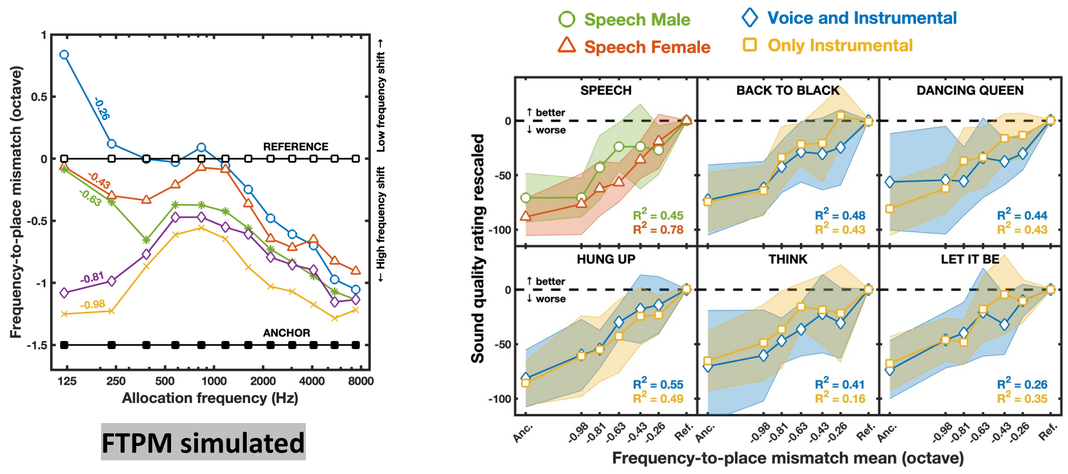
Speech & Music



Experiment 1 – Effect of frequency-to-place mismatch on sound quality

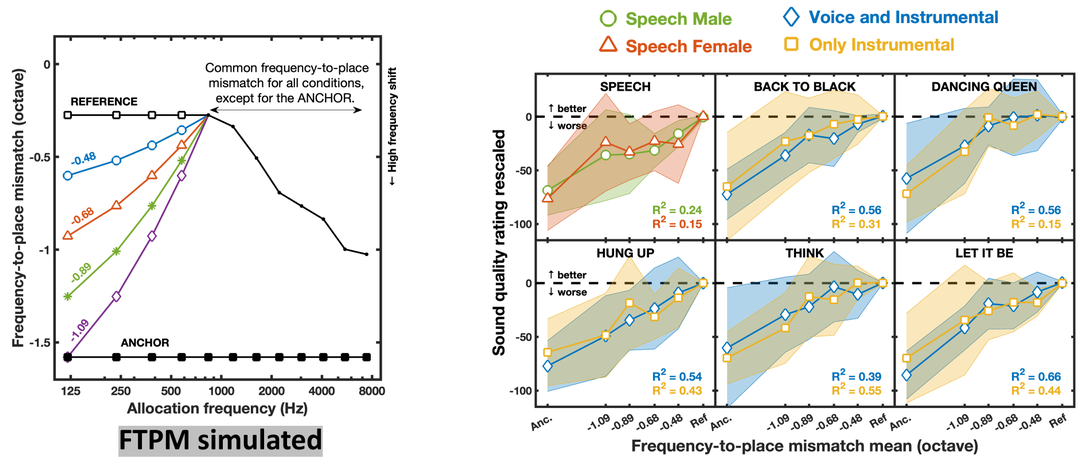
● Eight normal hearing listeners tested online ● FTPM simulated using a noise vocoder ● Speech, music and instrumental music tested ● MUSHRA method

Effect of realistic frequency-to-place mismatch



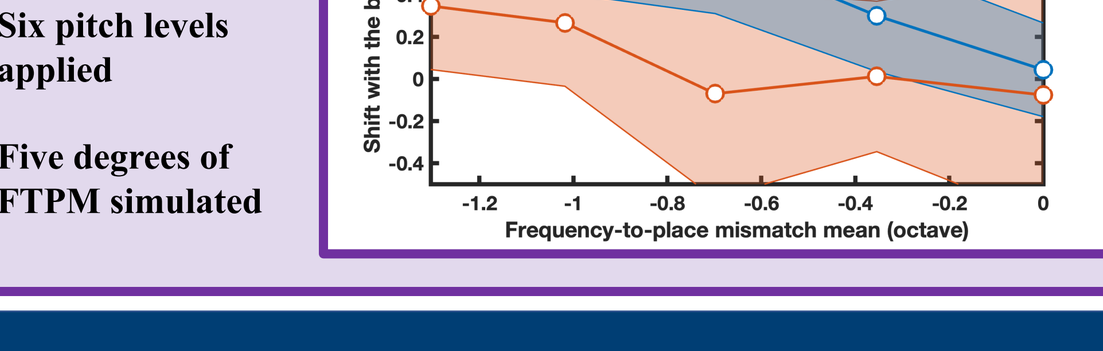
FTPM simulated

Effect of frequency-to-place mismatch in low frequencies



FTPM simulated

Effect of frequency-to-place mismatch in high frequencies



FTPM simulated

Results at a glance

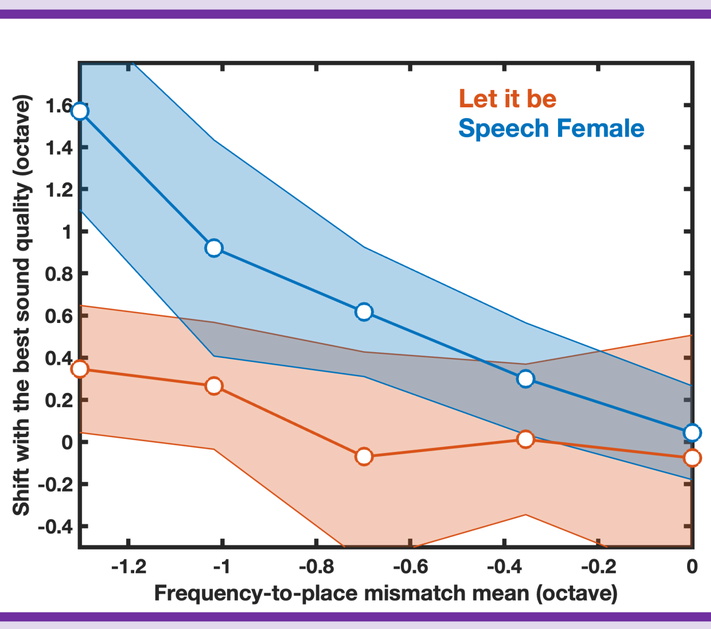
- ❑ As the frequency-to-place mismatch increases, the sound quality decreases.
- ❑ The quality of sound is more affected when the frequency- to-place mismatch is in the low frequencies rather than in high frequencies.
- ❑ Frequency-to-place mismatch affects sound materials differently.

Next step

- ❑ Understand why some sounds are more impacted than others.

Experiment 2 – Predicting the degree of frequency-to-place mismatch using pre-shifted sound

- Four normal hearing listeners tested
- Two sound materials tested
- Six pitch levels applied
- Five degrees of FTPM simulated



Results at a glance

- ❑ Based on our preliminary data, the speech used appears to be an effective predictor of the degree of FTPM simulated.
- ❑ At this stage, there is no clear trend regarding the effect of the music "Let it be”.

Next steps

- ❑ Increase the sample size by testing more participants.
- ❑ Expand the range of music tested.
- ❑ Improve the analysis methodology.
- ❑ Modify the MUSHRA to incorporate an adaptive approach that is appropriate for CI users.